LUMEXA
Space-themed coding education for K–12

AGES 6–12 | BEGINNER FRIENDLY

CURRICULUM OVERVIEW

Game Creator Path

Build 3 Real Games

A complete, project-driven journey from a first Roblox build to a polished, published game. Students don't just play games — they design them, script them, and ship three real, playable games friends can actually try.



3 REAL GAMES TO SHIP
Playable, shareable, portfolio-ready

4 TESTS ALONG THE WAY
3 course tests + 1 final test

3	24	28	6–12
Courses	Lessons	Sessions in total	Age range

WHO IT IS FOR
Beginner friendly — no experience needed

BEFORE YOU START
No coding experience needed

LEARNING PROGRESSION

01 Roblox World Builder
Build and publish a multiplayer Roblox game.

02 Python Arcade Games
Build 2D arcade games with Python and Pygame.

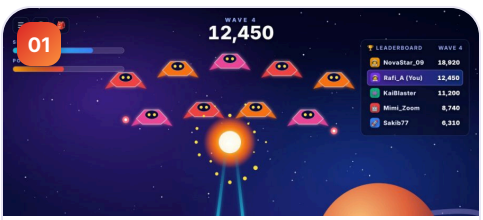
03 Advanced Game Design
Design, polish and publish a complete game.

Learn by shipping, not by watching
Every lesson pairs a real idea with a real, hands-on build. No shallow demos, no unfinished prototypes.

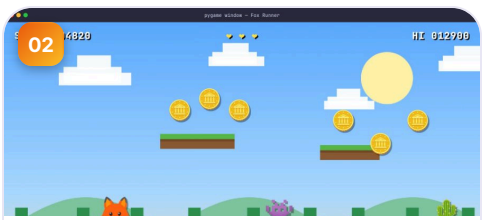
A portfolio, not just a certificate
Students finish with three real games they can publish, share and keep building on.

Tested, so progress is real
Every course ends with a test, and a final test closes the program, so progress is proven, not assumed.

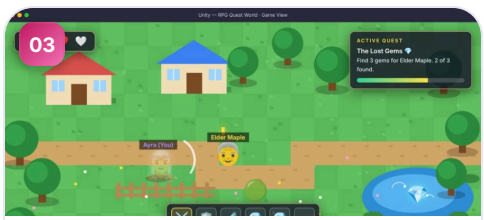
HANDS-ON PROJECTS STUDENTS BUILD



Galactic Shooter
Roblox · Lua



Python Arcade Game
Python · Pygame



RPG Quest World
Unity · C#

COURSE 01 · LESSONS 1–8

01 Roblox World Builder

Build and publish a multiplayer Roblox game from scratch

Roblox Studio | Lua scripting | Roblox DevHub

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Your child goes from zero to a working multiplayer Roblox game. They design the world, write the Lua code, set up game logic, and publish it so friends can actually play it.

PORTFOLIO PROJECTS

- Obstacle course with scoring
- Multiplayer arena with teams
- Custom game world with NPCs

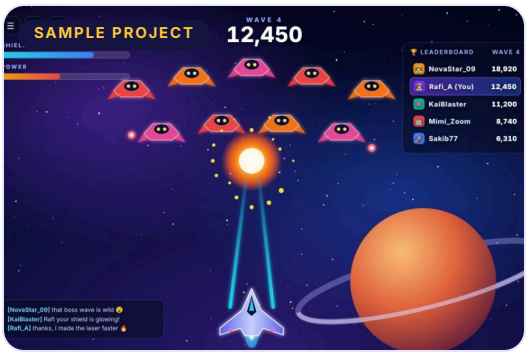
WHY IT MATTERS

Roblox is where millions of players already spend time — and where they build. Scripting in Lua teaches real programming logic while students create something their friends can join and play immediately.

LESSON BREAKDOWN

1. Roblox Studio Setup and First Build
2. Scripting Basics in Lua
3. Player Mechanics and Movement
4. Game Logic and Scoring Systems
5. Multiplayer Networking Basics
6. World Design and Aesthetics
7. Testing, Debugging, and Polishing
8. Publishing and Sharing with Friends

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

Galactic Shooter

Roblox · Lua

A space shooter with lasers, a score and a live leaderboard — the game loop and scoring ideas this course teaches.

COURSE TEST

Roblox World Builder — Course Test

Checks what you built and learned in Roblox World Builder. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

02

COURSE 02 · LESSONS 9–16

Python Arcade Games

Build 2D games using Python and Pygame

Python 3 | Pygame | VS Code

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

After Roblox, your child levels up to Python, the real language used by professional game developers and AI engineers. They build 2D arcade games from scratch.

PORTFOLIO PROJECTS

- Space shooter with enemies
- Maze runner with timer
- Brick-breaker clone

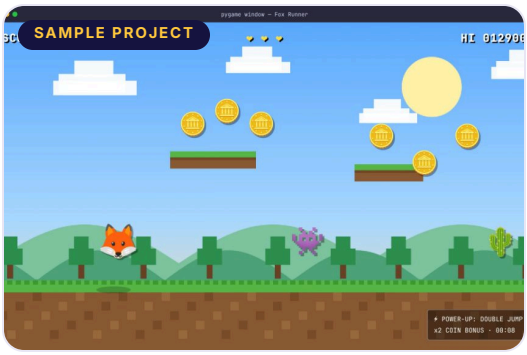
WHY IT MATTERS

Every real game — from indie hits to AAA titles — runs on the same fundamentals taught here: a game loop, sprites, collision detection, and state management. Pygame makes those ideas concrete and visible.

LESSON BREAKDOWN

1. Python Basics for Game Development
2. Pygame Setup and the Game Loop
3. Sprites, Images, and Movement
4. Collision Detection
5. Enemies, AI Behavior, and Scoring
6. Sound Effects and Music
7. Game States — Menu, Play, Game Over
8. Packaging and Sharing Your Game

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

Python Arcade Game

Python · Pygame

A side-scrolling arcade game with enemies, coins, power-ups and a high-score table, built with Pygame.

COURSE TEST

Python Arcade Games — Course Test

Checks what you built and learned in Python Arcade Games. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

COURSE 03 · LESSONS 17–24

03 Advanced Game Design

Design, polish, and publish a complete game with professional quality

Python / Pygame | Unity Basics | Game Design Frameworks

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

The capstone course. Your child applies everything they know to design and ship a complete game — proper level design, a polished UI, and real playtesters giving feedback.

PORTFOLIO PROJECTS

- RPG adventure with storyline
- Physics-based puzzle game
- Published portfolio game

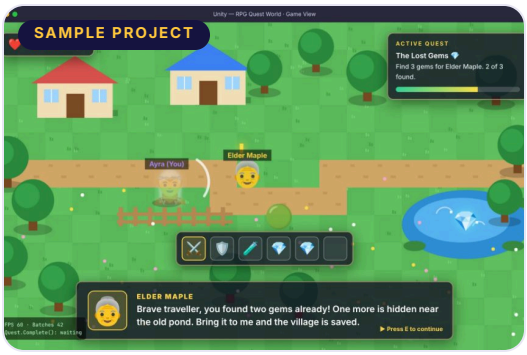
WHY IT MATTERS

Making a fun game takes more than code — it takes design thinking, iteration, and real player feedback. This course teaches the professional workflow behind every shipped title, from prototype to playtested, polished release.

LESSON BREAKDOWN

1. Game design fundamentals (loops, balance, flow)
2. Level design theory and practice
3. Advanced Python: classes and game architecture
4. Unity introduction and first scene
5. UI design and player experience
6. Playtesting and iteration
7. Performance optimization
8. Final polish and portfolio submission

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

RPG Quest World

Unity · C#

A top-down adventure with quests, friendly characters and simple combat — designed, playtested and polished.

COURSE TEST

Advanced Game Design — Course Test

Checks what you built and learned in Advanced Game Design. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

HOW STUDENTS ARE TESTED · 4 TESTS

Level-Up Checkpoints

Every game course ends with a course test, and a final test closes the path. They check the ideas behind the games — game loops, scripting, collisions and design — not just the finished build.



THE CHECKPOINT PATH — EVERY COURSE ENDS WITH A TEST, THEN ONE FINAL



COURSE TEST · after every 8 lessons

Show what you built and learned

After every 8 lessons a course test checks the ideas and skills of that course, so each next course starts on solid ground.

- **30** multiple-choice questions — choose one of four answers
- **10** true / false questions
- **40** marks in total — 1 mark for every question

FINAL TEST · the finish line

One final test on the whole curriculum

One final test across Roblox, Python games and game design: questions plus a small practical project.

- **25** multiple-choice questions
- **5** true / false questions
- **20** marks for one small practical project (about 15 minutes) that your teacher marks

A TASTE OF A TEST

SAMPLE · MULTIPLE-CHOICE

What does a game loop do on every frame?

- A It deletes the game
- B It updates the game and redraws everything ✓ CORRECT
- C It saves the high score online
- D It turns the sound off

SAMPLE · TRUE / FALSE

Collision detection checks whether two objects are touching.

True ✓ | False

Examples in the style of a real test — written for this document, not taken from one.

AFTER EVERY TEST

Saved as you go
Answers save automatically while a student works, so nothing is lost if the connection drops.

Checked by the teacher
The teacher reviews the report and shares the result with the student.

Fair for everyone
Questions and answer options appear in a different order for every student.

The Complete Journey

Across 3 courses, 24 lessons and 4 tests, students progress from their first Roblox build to a fully designed and polished game — walking away with three real, playable games for their portfolio.

3 COURSES · 24 LESSONS

01 Roblox World Builder
Obstacle course · Multiplayer arena · Custom world with NPCs
8 lessons · + course test

02 Python Arcade Games
Space shooter · Maze runner with timer · Brick-breaker clone
8 lessons · + course test

03 Advanced Game Design
RPG adventure · Physics puzzle game · Published portfolio game
8 lessons · + course test

HOW STUDENTS ARE TESTED · 4 TESTS

COURSE TEST 1
Roblox World Builder

COURSE TEST 2
Python Arcade Games

COURSE TEST 3
Advanced Game Design

FINAL TEST
One final test on the whole curriculum
One final test across Roblox, Python games and game design: questions plus a small practical project.

BY THE END, STUDENTS CAN

- Script and publish a working multiplayer game in Roblox Studio using Lua.
- Build a complete 2D arcade game in Python, including sprites, collisions, and sound.
- Design, playtest, and publish a polished, professional-quality game of their own.

Ready to start building?
All lessons and projects are taught live by an expert Lumexa instructor.

Lumexa Game Creator · Ages 6–12